Passage - 3

Soil, in which nearly all our food grows, is a living resource that takes years to form. Yet it can vanish in minutes. Each year 75 billion tonnes of fertile soil is lost to erosion. That is alarming----and not just for food producers. Soil can trap huge quantities of carbon dioxide in the form of organic carbon and prevent it from escaping into the atmosphere.

4. On the basis of the above passage, the following assumptions have been made:

1. Large scale soil erosion is a major reason for widespread food insecurity in the world.

2. Soil erosion is mainly anthropogenic.

3. Sustainable management of soils helps in combating climate change.

Which of the above assumptions is/are valid ?

(a) 1 and 2 only
(b) 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

Ans : (b) Assumption (1) is wrong as the link between soil erosion and food inrecurity is not established in the passage. The word alarming" does not mean that food insecurity has already arrived. Assumption (2) is incorrect. It is not mentioned anywhere in the passage that man is responsible for soil erosion. Assumption (3) in surely correct as borne out by the last line.

Passage - 4

Inequality is visible, even statistically measurable in many instances, but the economic power that drives it is invisible and not measurable. Like the force of gravity, power is the organising principle of inequality, be it of income, or wealth, gender, race, religion and region. Its effects are seen in a pervasive manner in all spheres, but the ways in which economic power pulls and tilts visible economic variables remain invisibly obscure.

5. On the basis of the above passage, the following assumptions have been made :

1. Economic power is the only reason for the existence of inequality in a society.

2. Inequality of different kinds, income, wealth, etc. reinforces power.

3. Economic power can be analysed more through its effects than by direct empirical methods.

Which of the above assumption is /are valid ?

(a) 1 and 2 only
(b) 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Ans : (b) On the basis of the above passage, the assumption (1), "economic power is the only reason for the existence of inequality in a society," cannot be the only reason, hence the assumption (1) is not valid. Assumption (2) is also the opposite when it is the forces that create and drive inequality. Hence, Assumption (2) is also wrong. Assumption (3) is valid because economic power can be analyzed through its effects rather than directly. Hence, Assumption (3) is valid. Hence option (b) is valid.

Passage - 5

Climate change may actually benefit some plants by lengthening growing seasons and increasing carbon dioxide. Yet other effects of a warmer world, such as more pests, droughts, and flooding, will be less benign. How will the world adapt? Researchers project that by 2050, suitable croplands for four commodities--maize, potatoes, rich and wheat--will shift, in some cases pushing farmers to plant new crops. Some farmlands may benefit from warming, but others won't . Climate alone does not dictate yields; political shifts, global demand, and agricultural practices will influence how farms fare in the future .

6. Which one of the following is the most logical and rational inference that can be made from the above passage

(a) Farmers who modernize their methods and diversify their fields will be in an advantageous position in future
(b) Climate change will adversely affect the crop diversity
(c) Shifting major crops to new croplands will lead to a great increase in the total area under cultivation and thus an increase in overall agricultural production.
(d) Climate change is the most important factor affecting the agricultural economy in the future.

Ands :(a) From the passage option (a) is correct inference. Farmers can be in an advantageous position by adopting new methods and diversifying their field.

Passage - 6

A bat's wings may look like sheets of skin. But underneath, a bat has the same five fingers as an orangutan or the human, as well as a wrist connected to the some cluster of wrist bones connected to the same long bones of the arm. What can be more curious than that the hand of a man, formed for grasping, that of the mole for digging, the led of the horse, the paddle of the porpoise, and the wing of the bat, should all be constructed on the same pattern?

7. Which one of the following is the *most logical, scientific and rational inference* that can be made from the above passage ?

(a) Different species having similar structure of hands is an example of biodiversity
(b) Limbs being used by different species for different kinds of work is an example of biodiversity
(c) Man and the aforementioned animals having similar structure of limbs is an example of coincidence in evolution.
(d) Man and the aforementioned animals have a shared evolutionary history.

Ans (d) : Option (a) is not mentioned specifically. Option (b) again talks about "biodiversity" but limbs being used for different purpose is not biodiversity. Option (c) sound correct, but is not, as "Coincidence in evolution" is not explicitly mentioned. Option (d) is best as "a shared evolutionary history" makes it a "very curious situation".

Passage - 7

Around 56 million years ago, the Atlantic Ocean had not fully opened and animals, perhaps including our primate ancestors, could walk from Asia to North America through Europe and across Greenland. Earth was warmer that it is today, but as the Paleocene epoch gave way to Eocene, it was about to get much warmer still-- rapidly and radically. The cause was a massive geologically sudden release of carbon. During this period called Paleocene-Eocene Thermal Maximum or PETM, the carbon injected into the atmosphere was roughly the amount that would be injected today if human burned all the Earth's reserves of coal, oil and natural gas. The PETM lasted for about 1,50,000 years, until the excess carbon was reabsorbed. It brought on drought, floods insect plagues and a few extinctions. Life on Earth survived--indeed, it prospered----but it was drastically different.

8. Based on the above passage, the following assumption have been made :

1. Global warming has a bearing on the planet's biological evolution.

2. Separation of land masses causes the release of huge quantities of carbon into the atmosphere.

3. Increased warming of Earth's atmosphere can change the composition of its flora and fauna.

4. The present man-made global warming will finally lead to conditions similar to those which happened 56 million years ago.

Which of the assumptions given above are valid

(a) 1 and 2 (b) 3 and 4
(c) 1 and 3 (d) 2 and 4

Ans : (c) Assumption (1) is clearly valid, Assumption (2) is too far-fetched and cannot be deduced. Assumption (3) is valid as clearly stated. Assumption (4) again is too far fetched as the amounts involved are too high.

Directions for the following 8 (eight) items :
Read the following six passage and answer the items that follow each passage. Your answer to these items should be based on the passages only.

Passage - 1

Low-end IoT (Internet of Things) devices are cheap commodity items : addressing security would add to the cost. This class of items is proliferating with new applications; many home appliances, thermostats, security and monitoring devices and personal convenience devices are part of the IoT. So are fitness trackers, certain medical implants and computer-like devices in automobiles. The IoT is expected to expand exponentially-but new security challenges are daunting.

9. Which one of the following statements is the *most logical and rational inference* that can be made from the above passage ?

(a) Development of enabling technologies in India can be a big boost to its manufacturing sector.
(b) India is not yet fully ready to adopt IoT in view of the imminent security challenges.
(c) Life becomes more comfortable with the development of cheap low-end IoT devices.
(d) As we go digital, we must recognise the huge threat to Internet security from some IoT devices.

Ans : (d) Option (a) is irrelevant as per the above passage. Option (b) is also irrelevant according to this passage. Because in the passage there was no mention of the matter in the context of India. Option (c) is not relevant in the context of the above paragraph. Option (d) is the most logical and prudent conclusion as it is warning about future dangers while interpreting the subjective passage.

Passage - 2

With the digital phenomenon restructuring most social sectors, it is little surprise that global trade negotiations are now eyeing the digital area in an attempt to pre-emptively colonise it. Big data is freely collected or mined from developing countries, and converted into digital intelligence in developed countries. This intelligence begging to control different sectors and extract monopoly rents. A large foreign company providing cab service, for instance, is not a network of cars and drivers ; it is digital intelligence about commuting, public transport, roads, traffic, city events, personal behavioural characteristics of commuters and drivers and so on.